# 配置新的 Cisco Nexus 92300YC 交换机

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## 配置新的 Cisco Nexus 92300YC 交换机

### 您可以通过完成本章中详细介绍的步骤来配置新的 Nexus 92300YC 交换机。

在运行 ONTAP 9.6 及更高版本的系统上安装 Nexus 92300YC 交换机时，首先需要设置 IP 地址和配置，以使交换机能够通过管理接口进行通信。然后，您可以安装 NX-OS 软件和参考配置文件（RCF）。此操作步骤用于在添加控制器之前准备 Nexus 92300YC 交换机。

此操作步骤中的示例使用以下交换机和节点命名：

- Nexus 92300YC 交换机名称为 `CS1` 和 `CS2` 。
- 此操作步骤中使用的示例将在第二个交换机 _* CS2* 上启动升级。 _
- 集群 LIF 名称是 node1 的 `node1_clus1` 和 `node1_clus2` ， node2 的 `node2_clus1` 和 `node2_clus2` 。
- IP 空间名称是 `Cluster` 。
- `cluster1 ：：： * >` 提示符指示集群的名称。
- 每个节点上的集群端口分别命名为 `e0a` 和 `e0b` 。

  有关 Hardware Universe 您的平台支持的实际集群端口，请参见 *cluster* 。

  "NetApp Hardware Universe"

- Nexus 92300YC 交换机支持的交换机间链路（ISL）为端口 1/65 和 1/66 。
- Nexus 92300YC 交换机支持的节点连接为端口 1/1 到 1/66 。
- 此操作步骤中的示例使用两个节点，但一个集群中最多可以有 24 个节点。

## 首次安装 Cisco Nexus 92300YC 交换机

### 您可以使用此操作步骤执行 Cisco Nexus 92300YC 交换机的初始安装。

您可以从 NetApp 支持站点下载适用于您的交换机的 NetApp Cisco NX-OS 软件，网址为 "mysupport.netapp.com"

NX-OS 是 Cisco Systems 提供的 Nexus 系列以太网交换机和 MDS 系列光纤通道（FC）存储区域网络交换机的网络操作系统。

此操作步骤简要介绍了安装交换机并使其运行的过程：

步骤

1. 将串行端口连接到所选的主机或串行端口。
2. 将管理端口（位于交换机的非端口端）连接到您的 SFTP 服务器所在的同一网络。
3. 在控制台中，设置主机端串行设置：

- 9600 波特
- 8 个数据位
- 1 个停止位
- 奇偶校验：无
- 流量控制：无

4. Nexus 92300YC 交换机在首次启动或擦除正在运行的配置后重新启动时会在启动周期中循环。键入 * 是 * 以中止启动自动配置，从而中断此周期。然后，系统将显示 System Admin Account 设置：

```
$ VDC-1 %$ %POAP-2-POAP_INFO:   - Abort Power On Auto Provisioning [yes
- continue with normal setup, skip - bypass password and basic
configuration, no - continue with Power On Auto Provisioning]
(yes/skip/no)[no]:
**yes**
Disabling POAP.......Disabling POAP
2019 Apr 10 00:36:17 switch %$ VDC-1 %$ poap: Rolling back, please
wait... (This may take 5-15 minutes)
```

```
---- System Admin Account Setup ----
```

```
 Do you want to enforce secure password standard (yes/no) [y]:
. 键入 * y* 以强制实施安全密码标准：
```

```
Do you want to enforce secure password standard (yes/no) [y]: **y**
```

5. 输入并确认用户 admin 的密码：

```
Enter the password for "admin":
Confirm the password for "admin":
```

6. 进入基本系统配置对话框：

```
This setup utility will guide you through the basic configuration of
the system. Setup configures only enough connectivity for management
of the system.

Please register Cisco Nexus9000 Family devices promptly with your
supplier. Failure to register may affect response times for initial
service calls. Nexus9000 devices must be registered to receive
entitled support services.

Press Enter at anytime to skip a dialog. Use ctrl-c at anytime
to skip the remaining dialogs.

Would you like to enter the basic configuration dialog (yes/no):
```

7. 是否创建其他登录帐户?

```
Create another login account (yes/no) [n]:
```

8. 配置只读和读写 SNMP 社区字符串:

```
Configure read-only SNMP community string (yes/no) [n]:

Configure read-write SNMP community string (yes/no) [n]:
```

9. 配置集群交换机名称:

```
Enter the switch name : **cs2**
```

10. 配置带外管理接口:

```
Continue with Out-of-band (mgmt0) management configuration? (yes/no)
[y]: **y**

Mgmt0 IPv4 address : 172.22.133.216

Mgmt0 IPv4 netmask : 255.255.224.0

Configure the default gateway? (yes/no) [y]: **y**

IPv4 address of the default gateway : 172.22.128.1
```

11. 是否配置高级 IP 选项?

```
Configure advanced IP options? (yes/no) [n]: **n**
```

12. 配置 Telnet 服务:

```
Enable the telnet service? (yes/no) [n]: **n**
```

13. 配置 SSH 服务和 SSH 密钥:

```
Enable the ssh service? (yes/no) [y]: **y**

       Type of ssh key you would like to generate (dsa/rsa) [rsa]:
**rsa**

       Number of rsa key bits <1024-2048> [1024]: **2048**
```

14. 配置其他设置:

```
Configure the ntp server? (yes/no) [n]: **n**

  Configure default interface layer (L3/L2) [L2]: **L2**

  Configure default switchport interface state (shut/noshut) [noshut]:
**noshut**

  Configure CoPP system profile (strict/moderate/lenient/dense)
[strict]: **strict**
```

15. 确认交换机信息并保存配置:

```
Would you like to edit the configuration? (yes/no) [n]: **n**

Use this configuration and save it? (yes/no) [y]: **y**

[########################################] 100%
Copy complete, now saving to disk (please wait)...
Copy complete.
```

# 安装 NX-OS 软件

您可以使用此操作步骤在 Nexus 92300YC 交换机上安装 NX-OS 软件。

步骤

1. 将集群交换机连接到管理网络。
2. 使用 `ping` 命令验证与托管 NX-OS 软件和 RCF 的服务器的连接。

   此示例验证交换机是否可以通过 IP 地址 172.19.2.1 访问服务器：

   ```
   cs2# **ping 172.19.2.1**
   Pinging 172.19.2.1 with 0 bytes of data:

   Reply From 172.19.2.1: icmp_seq = 0. time= 5910 usec.
   ```

3. 将 NX-OS 软件和 EPLD 映像复制到 Nexus 92300YC 交换机。

```
cs2# **copy sftp: bootflash: vrf management**
Enter source filename: **/code/nxos.9.2.2.bin**
Enter hostname for the sftp server: **172.19.2.1**
Enter username: **user1**

Outbound-ReKey for 172.19.2.1:22
Inbound-ReKey for 172.19.2.1:22
user1@172.19.2.1's password:
sftp> progress
Progress meter enabled
sftp> get   /code/nxos.9.2.2.bin  /bootflash/nxos.9.2.2.bin
/code/nxos.9.2.2.bin  100% 1261MB   9.3MB/s   02:15
sftp> exit
Copy complete, now saving to disk (please wait)...
Copy complete.


cs2# **copy sftp: bootflash: vrf management**
Enter source filename: **/code/n9000-epld.9.2.2.img**
Enter hostname for the sftp server: **172.19.2.1**
Enter username: **user1**

Outbound-ReKey for 172.19.2.1:22
Inbound-ReKey for 172.19.2.1:22
user1@172.19.2.1's password:
sftp> progress
Progress meter enabled
sftp> get   /code/n9000-epld.9.2.2.img  /bootflash/n9000-epld.9.2.2.img
/code/n9000-epld.9.2.2.img  100%  161MB   9.5MB/s   00:16
sftp> exit
Copy complete, now saving to disk (please wait)...
Copy complete.
```

4. 验证正在运行的 NX-OS 软件版本：

```
cs2# **show version**
Cisco Nexus Operating System (NX-OS) Software
TAC support: http://www.cisco.com/tac
Copyright (C) 2002-2018, Cisco and/or its affiliates.
All rights reserved.
The copyrights to certain works contained in this software are
owned by other third parties and used and distributed under their own
licenses, such as open source.  This software is provided "as is," and
unless
otherwise stated, there is no warranty, express or implied, including
```

```
but not
limited to warranties of merchantability and fitness for a particular
purpose.
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http://www.opensource.org/licenses/lgpl-2.1.php and
http://www.gnu.org/licenses/old-licenses/library.txt.

Software
  BIOS: version 05.31
  NXOS: version 9.2(1)
  BIOS compile time:  05/17/2018
  NXOS image file is: bootflash:///nxos.9.2.1.bin
  NXOS compile time:  7/17/2018 16:00:00 [07/18/2018 00:21:19]


Hardware
  cisco Nexus9000 C92300YC Chassis
  Intel(R) Xeon(R) CPU D-1526 @ 1.80GHz with 16337884 kB of memory.
  Processor Board ID FDO220329V5

  Device name: cs2
  bootflash:  115805356 kB
Kernel uptime is 0 day(s), 4 hour(s), 23 minute(s), 11 second(s)

Last reset at 271444 usecs after Wed Apr 10 00:25:32 2019
  Reason: Reset Requested by CLI command reload
  System version: 9.2(1)
  Service:

plugin
  Core Plugin, Ethernet Plugin

Active Package(s):

cs2#
```

5. 安装 NX-OS 映像。

   安装映像文件会导致每次重新启动交换机时加载该映像文件。

```
cs2# **install all nxos bootflash:nxos.9.2.2.bin**

Installer will perform compatibility check first. Please wait.
Installer is forced disruptive

Verifying image bootflash:/nxos.9.2.2.bin for boot variable "nxos".
[####################] 100% -- SUCCESS

Verifying image type.
[####################] 100% -- SUCCESS

Preparing "nxos" version info using image bootflash:/nxos.9.2.2.bin.
[####################] 100% -- SUCCESS

Preparing "bios" version info using image bootflash:/nxos.9.2.2.bin.
[####################] 100% -- SUCCESS

Performing module support checks.
[####################] 100% -- SUCCESS

Notifying services about system upgrade.
[####################] 100% -- SUCCESS



Compatibility check is done:
Module  bootable       Impact     Install-type  Reason
------  --------  --------------- ------------  ------
  1       yes      disruptive         reset     default upgrade is not
hitless



Images will be upgraded according to following table:

Module   Image         Running-Version(pri:alt            New-Version
Upg-Required
------  --------  --------------------------------------
------------------  ------------
  1       nxos                                  9.2(1)
9.2(2)         yes
  1       bios    v05.31(05/17/2018):v05.28(01/18/2018)
v05.33(09/08/2018)       yes


Switch will be reloaded for disruptive upgrade.
```

```
Do you want to continue with the installation (y/n)?  [n] **y**


Install is in progress, please wait.

Performing runtime checks.
[####################] 100% -- SUCCESS

Setting boot variables.
[####################] 100% -- SUCCESS

Performing configuration copy.
[####################] 100% -- SUCCESS

Module 1: Refreshing compact flash and upgrading bios/loader/bootrom.
Warning: please do not remove or power off the module at this time.
[####################] 100% -- SUCCESS
2019 Apr 10 04:59:35 cs2 %$ VDC-1 %$ %VMAN-2-ACTIVATION_STATE:
Successfully deactivated virtual service 'guestshell+'

Finishing the upgrade, switch will reboot in 10 seconds.
```

6. 在交换机重新启动后验证 NX-OS 软件的新版本：

   s如何使用版本

```
cs2# **show version**

Cisco Nexus Operating System (NX-OS) Software
TAC support: http://www.cisco.com/tac
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licenses, such as open source.  This software is provided "as is," and
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GNU General Public License (GPL) version 3.0  or the GNU
Lesser General Public License (LGPL) Version 2.1 or
Lesser General Public License (LGPL) Version 2.0.
A copy of each such license is available at
```

```
http://www.opensource.org/licenses/gpl-2.0.php and
http://opensource.org/licenses/gpl-3.0.html and
http://www.opensource.org/licenses/lgpl-2.1.php and
http://www.gnu.org/licenses/old-licenses/library.txt.

Software
  BIOS: version 05.33
  NXOS: version 9.2(2)
  BIOS compile time:  09/08/2018
  NXOS image file is: bootflash:///nxos.9.2.2.bin
  NXOS compile time:  11/4/2018 21:00:00 [11/05/2018 06:11:06]


Hardware
  cisco Nexus9000 C92300YC Chassis
  Intel(R) Xeon(R) CPU D-1526 @ 1.80GHz with 16337884 kB of memory.
  Processor Board ID FDO220329V5

  Device name: cs2
  bootflash:  115805356 kB
  Kernel uptime is 0 day(s), 0 hour(s), 3 minute(s), 52 second(s)

Last reset at 182004 usecs after Wed Apr 10 04:59:48 2019
  Reason: Reset due to upgrade
  System version: 9.2(1)
  Service:

plugin
  Core Plugin, Ethernet Plugin

Active Package(s):
```

7. 升级 EPLD 映像并重新启动交换机。

```
cs2# **show version module 1 epld**

EPLD Device                      Version
---------------------------------------
MI FPGA                           0x7
IO FPGA                           0x17
MI FPGA2                          0x2
GEM FPGA                          0x2
GEM FPGA                          0x2
GEM FPGA                          0x2
GEM FPGA                          0x2

cs2# **install epld bootflash:n9000-epld.9.2.2.img module 1**
Compatibility check:
Module        Type         Upgradable        Impact   Reason
------  -----------------  ----------    ----------   ------
     1            SUP           Yes       disruptive  Module Upgradable

Retrieving EPLD versions.... Please wait.
Images will be upgraded according to following table:
Module  Type   EPLD              Running-Version   New-Version  Upg-
Required
------  ----  -------------      ---------------   -----------
------------
     1   SUP  MI FPGA                   0x07        0x07             No
     1   SUP  IO FPGA                   0x17        0x19            Yes
     1   SUP  MI FPGA2                  0x02        0x02             No
The above modules require upgrade.
The switch will be reloaded at the end of the upgrade
Do you want to continue (y/n) ?  [n] **y**

Proceeding to upgrade Modules.

Starting Module 1 EPLD Upgrade

Module 1 : IO FPGA [Programming] : 100.00% (     64 of      64 sectors)
Module 1 EPLD upgrade is successful.
Module        Type  Upgrade-Result
------  ------------------  --------------
     1         SUP         Success


EPLDs upgraded.

Module 1 EPLD upgrade is successful.
```

8. 交换机重新启动后，重新登录并验证是否已成功加载新版本的 EPLD 。

```
cs2# **show version module 1 epld**

EPLD Device                      Version
---------------------------------------
MI FPGA                           0x7
IO FPGA                           0x19
MI FPGA2                          0x2
GEM FPGA                          0x2
GEM FPGA                          0x2
GEM FPGA                          0x2
GEM FPGA                          0x2
```

# 安装参考配置文件（RCF）

首次设置 Nexus 92300YC 交换机后，您可以安装 RCF 。

步骤

1. 将集群交换机连接到管理网络。
2. 使用 ping 命令验证与托管 RCF 的服务器的连接。

   此示例验证交换机是否可以通过 IP 地址 172.19.2.1 访问服务器：

```
cs2# **ping 172.19.2.1**
Pinging 172.19.2.1 with 0 bytes of data:

Reply From 172.19.2.1: icmp_seq = 0. time= 5910 usec.
```

3. 将 RCF 复制到 Nexus 92300YC 交换机：

```
cs2# **copy sftp: bootflash: vrf management**
Enter source filename: **/code/Nexus_92300YC_RCF_v1.0.2.txt**
Enter hostname for the sftp server: **172.19.2.1**
Enter username: **user1**

Outbound-ReKey for 172.19.2.1:22
Inbound-ReKey for 172.19.2.1:22
user1@172.19.2.1's password:
sftp> progress
Progress meter enabled
sftp> get   /code/Nexus_92300YC_RCF_v1.0.2.txt
/bootflash/nxos.9.2.2.bin
/code/Nexus_92300YC_R  100% 9687   530.2KB/s   00:00
sftp> exit
Copy complete, now saving to disk (please wait)...
Copy complete.
```

4. 将 RCF 与交换机的运行配置合并：

```
cs2# **copy bootflash:Nexus_92300YC_RCF_v1.0.2.txt running-config**

Disabling ssh: as its enabled right now:
 generating ecdsa key(521 bits)......
generated ecdsa key

Enabling ssh: as it has been disabled
 this command enables edge port type (portfast) by default on all
interfaces. You
 should now disable edge port type (portfast) explicitly on switched
ports leading to hubs,
 switches and bridges as they may create temporary bridging loops.

Edge port type (portfast) should only be enabled on ports connected to a
single
 host. Connecting hubs, concentrators, switches, bridges, etc...  to
this
 interface when edge port type (portfast) is enabled, can cause
temporary bridging loops.
 Use with CAUTION

Edge Port Type (Portfast) has been configured on Ethernet1/1 but will
only
 have effect when the interface is in a non-trunking mode.

...

Copy complete, now saving to disk (please wait)...
Copy complete.
```

5. 在交换机上验证 RCF 是否已成功合并：

   s如何运行配置

```
cs2# **show running-config**
!Command: show running-config
!Running configuration last done at: Wed Apr 10 06:32:27 2019
!Time: Wed Apr 10 06:36:00 2019

version 9.2(2) Bios:version 05.33
switchname cs2
vdc cs2 id 1
  limit-resource vlan minimum 16 maximum 4094
  limit-resource vrf minimum 2 maximum 4096
  limit-resource port-channel minimum 0 maximum 511
  limit-resource u4route-mem minimum 248 maximum 248
  limit-resource u6route-mem minimum 96 maximum 96
  limit-resource m4route-mem minimum 58 maximum 58
  limit-resource m6route-mem minimum 8 maximum 8

feature lacp

no password strength-check
username admin password 5
$5$HY9Kk3F9$YdCZ8iQJ1RtoiEFa0sKP5IO/LNG1k9C4lSJfi5kesl
6  role network-admin
ssh key ecdsa 521

banner motd #
***********************************************************************
********
*
*
*  Nexus 92300YC Reference Configuration File (RCF) v1.0.2 (10-19-2018)
*
*
*
*  Ports 1/1  - 1/48: 10GbE Intra-Cluster Node Ports
*
*  Ports 1/49 - 1/64: 40/100GbE Intra-Cluster Node Ports
*
*  Ports 1/65 - 1/66: 40/100GbE Intra-Cluster ISL Ports
*
*
*
***********************************************************************
********
```

6. 保存正在运行的配置，使其在重新启动交换机时成为启动配置：

```
cs2# **copy running-config startup-config**

[#########################################] 100%
Copy complete, now saving to disk (please wait)...
Copy complete.
```

7. 对于 ONTAP 9.6P8 及更高版本，启用 CSHM ASUP 日志收集功能以收集交换机相关的日志文件：`ssystem cluster-switch log setup-password` 和 `ssystem cluster-switch log enable-Collection`

```
cs2# **system cluster-switch log setup-password**
** Output example required here
cs2# **system cluster-switch log enable-collection**
** Output example required here too
```

8. 重新启动交换机并验证正在运行的配置是否正确：

   `re`负载

```
cs2# **reload**

This command will reboot the system. (y/n)?  [n] **y**
```

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